

MVSR Engineering College
Department of Computer Science & Engineering-Allied
Academic Year: 2026-2027

B. E CSE (AI & ML) - V Semester Mini Based Project Acceptance Form

Date: 19/09/2026

1. Batch No:04

2. Mini Project Batch Details:

Name of the Student	Roll No.	Ph. No.	Signature
D.Varshith Reddy	2451-24-748-014	+91 90598 45457	
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3. Title of the Project: Sovereign Agentic AI Workbench for Secure Multimodal Industrial Data Intelligence

4. Domain: Artificial Intelligence

5. Abstract: (Minimum 100 words)

Industrial organizations generate large volumes of confidential and multimodal data, including technical documents, maintenance reports, machine images, sensor logs, and audio records. Sending such data to external cloud-based AI services can create risks related to data leakage, intellectual property protection, and regulatory compliance. This project proposes a Sovereign Agentic AI Workbench that enables intelligent processing and analysis of sensitive industrial data within a controlled local environment. The system uses open-weight multimodal Large Language Models (LLMs) deployed locally to understand text, images, documents, and structured sensor data. An agentic architecture allows specialized AI agents to retrieve relevant information, perform data analysis, use authorized tools, and generate evidence-based reports. A private Retrieval-Augmented Generation (RAG) pipeline provides access to organization-specific knowledge without exposing it externally. Security mechanisms such as role-based access control, encryption, audit logging, sandboxed tool execution, and network isolation are incorporated to protect confidential information. The complete platform is containerized using Docker to provide portability, reproducibility, and scalable deployment. The proposed system aims to demonstrate how open-weight multimodal AI can provide autonomous industrial intelligence while maintaining data sovereignty and human oversight.

6. Tools/Libraries/API's used: Python, FastAPI, React, Ollama, vLLM, LangGraph, LlamaIndex, Qdrant, PostgreSQL, Pandas, NumPy, Scikit-learn, PyMuPDF, OpenCV, Pillow, Matplotlib, Plotly, JWT, REST API, Docker, Docker Compose, Git, GitHub

Signature of Internal Guide

Project Approved / Not Approved
Remarks:

Project Review Committee Signature

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